

Week 4, Lecture 1: Bestiary of Distributions

Instructor notes – Part 1: prep reading, Part 2: podium cheat sheet

This single file has two parts, both instructor-only (not a student handout). **Part 1** is read beforehand – full worked examples and the reasoning behind the flow. **Part 2** (after the page break, large print) is the podium copy.

Source material: Bolker 4.4-4.6 (p. 155-188): analyzing probability distributions (moments), the bestiary of distributions, and extending simple distributions. Also mined: `LECTURE2.Rmd` (legacy version of this lecture pair – its Poisson/binomial/beta R demos are reused below with fresh parameter values), and the Week 3 Lecture 2 notes.

Where this sits in the week

Reminder of the Week 3/4 seam: Labor Day cancelled Week 3's Monday meeting, so Week 3 had only Lab 3 (Tuesday) and one lecture (Wednesday: probability foundations, Bayes' rule) – this bestiary lecture, which would normally have followed right behind on the same Wednesday, instead becomes Week 4's opening lecture.

- Last Tuesday (Lab 3): entirely pen-and-paper, no laptops. Act 1 derived the PPV/Bayes' rule calculation from plain counting (before any lecture on it); Acts 2-3 hand-computed binomial pmf values, read pre-rendered binomial/Poisson comparison figures, and used a printed z-table for the normal distribution – so students already know the *math* behind binomial, Poisson, and normal, but have never typed `dbinom()`, `dpois()`, `dnorm()`, `pnorm()`, or `qnorm()` themselves.
- Last Wednesday: probability foundations, Bayes' rule (formalized what Lab 3 had already had students derive).
- **Today (Week 4, Monday):** name what the lab already computed by hand, add moments and two distributions the lab didn't cover (a skewed continuous one, and a quick look at what happens when simple distributions don't fit), give students a framework for choosing a distribution from the shape of their own data – and, since the lab was laptop-free, this is students' *actual* first look at the `d/p/q/r` R syntax for these distributions, not just a review.
- **This Wednesday (Week 4, Lecture 2):** synthetic data simulation – today's distributions become the stochastic half of a full data-generating model.

Rationale for the flow

Bolker's own advice for section 4.5: skim it as a map, not a memorization list. Students already worked out the *math* of the binomial and Poisson pmf by hand in last week's lab, and used a z-table for the normal – today isn't a first introduction to the concepts, but it *is* the first time they'll see `dbinom()/dpois()/dnorm()` actually typed and run, so don't rush past that syntax reveal the way you might have if the lab itself had used it. Organize what they already know conceptually, connect it explicitly to the R functions that compute it, and add vocabulary (moments, support, the data-generating-process framing) plus 1-2 new shapes. Keep the survey shallow and wide, with one deeper worked example (moments) and one genuinely new distribution (a bounded continuous one – beta – since nothing in the lab needed it).

Part A: Moments (~8-10 min)

Moments describe the shape of a distribution. The n th moment is $E[X^n]$; the n th *central* moment replaces X with $X - \mu$.

- 0th central moment: always 1 (axiom 2, restated)
- 1st central moment: always 0 (that's what makes μ the mean)
- 2nd central moment: the **variance**
- 3rd, standardized: **skewness**; 4th, standardized: **kurtosis**

Worked derivation (Calc-II algebra only, no new calculus): show that $\text{Var}(X) = E[X^2] - \mu^2$, a formula students will use constantly but rarely see derived.

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2$$

Since $E[X] = \mu$:

$$= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$$

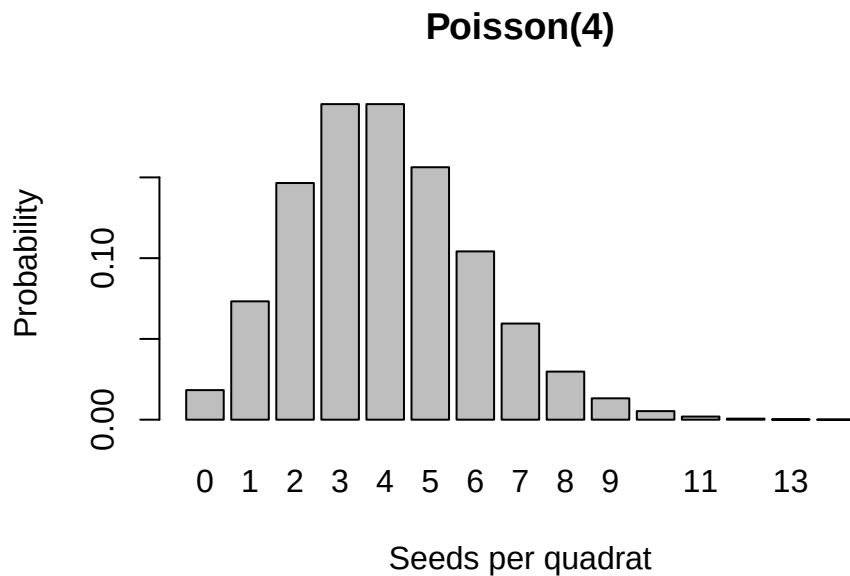
Worth pointing out explicitly: this is just expanding a square and using linearity of expectation – nothing here is harder than what Week 1's algorithms lecture already assumed they could do.

Parameters vs. moments: parameters are the numbers that go into a distribution's formula (e.g. λ for Poisson); moments are properties you can compute *from* the distribution once you know its parameters. Sometimes they're the same thing (Poisson's λ is simultaneously its mean *and* its variance) and sometimes they're not (the normal's σ^2 is its variance directly, but the gamma's variance is a function of two separate parameters).

Part B: Discrete distributions – counts (~8 min)

Fresh example, not last week's camera-trap detections: seed counts per quadrat. Suppose the average number of a given plant's seeds found per 1m^2 quadrat is $\lambda = 4$.

```
lambda <- 4
xvals <- 0:14
probs <- dpois(xvals, lambda)
barplot(probs, names.arg = xvals, ylab = "Probability",
        xlab = "Seeds per quadrat", main = "Poisson(4)")
```



```
sum(probs)      # doesn't quite hit 1 -- tail beyond x=14 is still out there
```

```
## [1] 0.9999801
```

Binomial recap (already met in Lab 3 Act 2): counts of successes out of a fixed number of independent trials, each with the same success probability p . **Poisson recap (also met in lab):** counts of independent events in a fixed window of time or space, with rate λ – the limiting case of the binomial as $N \rightarrow \infty$, $p \rightarrow 0$, with $Np = \lambda$ held fixed (this is exactly what Act 2's $N = 100$, $p = 0.024$ comparison demonstrated numerically).

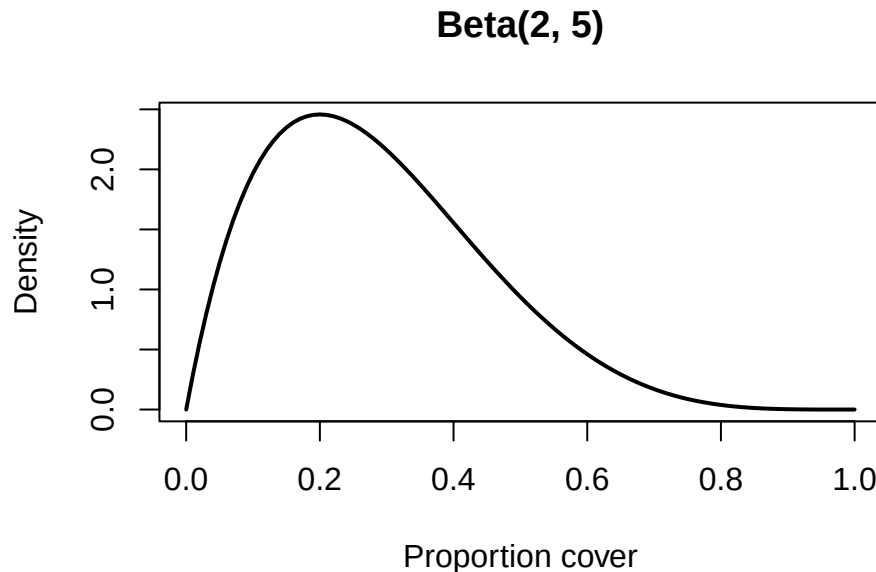
Sticking point to flag: Poisson's single parameter λ is *simultaneously* the mean and the variance (mean = variance is called **equidispersion**). Real count data is very often *more* variable than this (**overdispersion**) – worth a one-sentence preview that the negative binomial exists specifically to relax this constraint (not covered today).

Part C: Continuous distributions – measurements (~8-10 min)

Normal recap (met in Lab 3 Act 3): symmetric, unbounded, fully determined by its first two moments (μ , σ^2) – central to the whole course because of the central limit theorem.

New today – the **beta distribution**, for data that’s a *proportion* or *probability* itself (bounded between 0 and 1). Example: fraction of a quadrat’s area covered by an invasive species.

```
a <- 2; b <- 5    # shape parameters
curve(dbeta(x, a, b), from = 0, to = 1, lwd = 2,
      xlab = "Proportion cover", ylab = "Density", main = "Beta(2, 5)")
```



```
integrate(dbeta, lower = 0, upper = 1, shape1 = a, shape2 = b)    # check: integrates to
```

```
## 1 with absolute error < 1.1e-14
```

Point out: beta’s two shape parameters (α , β) don’t map onto mean/variance as directly as the normal’s do, but the mean works out to $\alpha/(\alpha + \beta)$ – worth showing since it’s exactly the “fraction of total” intuition the shape parameters suggest. This is also the natural distribution for *a probability itself* – worth a one-sentence flag forward to the Bayesian inference weeks, where a prior on a probability parameter is very often a beta.

Sticking point to flag (recap from Lab 3 Act 3, worth restating with a new example): the height of `dbeta()`’s curve is a *density*, not a probability – it can exceed 1 (try `dbeta(0.2, 2, 5)`), and only an *area* under the curve is a probability.

Part D: Choosing a distribution (~6-8 min)

The organizing question, straight from Bolker: **what process generated this data?**, not “what shape does the histogram look like?” A table worth putting on the board:

Data looks like...	Consider...
successes out of a fixed number of trials	binomial
counts of independent events, no fixed upper bound	Poisson (or negative binomial, if overdispersed)
a bounded proportion or probability	beta
a continuous, roughly symmetric measurement	normal
a continuous, right-skewed, positive measurement	gamma or lognormal (not covered in depth today)

The two “not covered in depth today” mentions (negative binomial, gamma/lognormal) are deliberate – flag them as *named* so students recognize them later in the semester, without deriving them now.

Take-home message

- Moments describe shape; $\text{Var}(X) = E[X^2] - \mu^2$ is worth remembering, not just looking up.
- Distributions are chosen by the *mechanism* that generated the data, not by eyeballing a histogram shape.
- Density $\neq$ probability (only area under a density curve is a probability); Poisson’s mean = variance is a real, checkable assumption, not just a mathematical curiosity.

Roadmap tie-in

- The shapes surveyed today (Weeks 1-2’s deterministic functions) will soon get noise wrapped around them – this Wednesday (synthetic data simulation, later this same week) combines a deterministic mean function with one of today’s distributions to build a full stochastic model.
- Overdispersion (Poisson’s broken equidispersion assumption) comes back explicitly in Week 11 (modeling variance).

Discussion seed (if time remains)

Ask: *looking back at last week’s lab – which distribution from today’s tour best describes the camera-trap detection counts (Act 2)? The body mass measurements (Act 3)? Can you imagine a way reality might violate the distribution you picked (e.g., detections clustering in time rather than being independent)?*

PART 2 – PODIUM CHEAT SHEET

Podium copy – terse on purpose. Full derivations, rationale, and code: see Part 1 above.

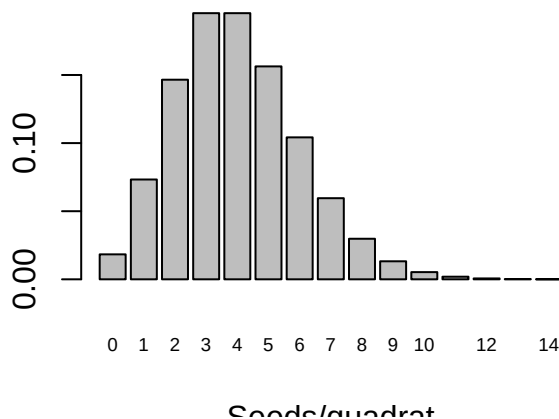
0. Open (~2 min)

- Last week’s lab hand-computed this math already (no laptops) – today is your first look at `dbinom()`/`dpois()`/`dnorm()` themselves, plus moments + one new shape

1. Moments (~8-10 min)

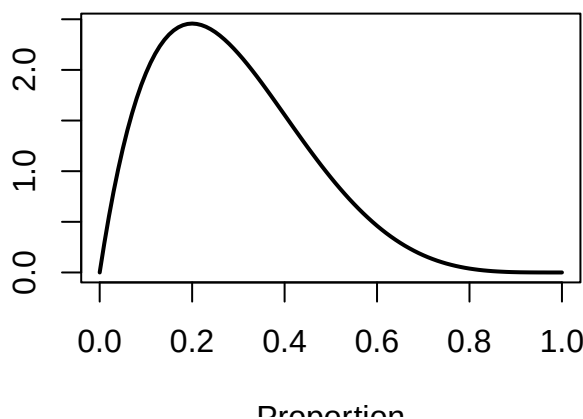
- 0th central moment = 1, 1st = 0, 2nd = **variance**, 3rd = skewness, 4th = kurtosis
- **Derive on board:** $\text{Var}(X) = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2$ (just expand the square + linearity of $E[\cdot]$)
- parameters (go into the formula) vs. moments (computed from it) – Poisson’s λ is *both* at once

2. Discrete: counts (~8 min)



- **Binomial** (Lab 3 Act 2): fixed N trials, constant p
- **Poisson** (Lab 3 Act 2): rate λ , limit of binomial as $N \rightarrow \infty, p \rightarrow 0, Np = \lambda$
- **Sticking point:** Poisson mean = variance (**equidispersion**) – real counts are often **overdispersed** (negative binomial handles this, not today)

3. Continuous: measurements (~8-10 min)



- **Normal** (Lab 3 Act 3): symmetric, unbounded, μ, σ^2 fully determine it (CLT)
- **New – Beta**: bounded $[0, 1]$, for proportions/probabilities; mean = $\alpha/(\alpha + \beta)$; natural prior for a probability parameter (preview: Bayesian weeks)
- **Sticking point (again)**: density $\neq$ probability – can exceed 1, only *area* is a probability

4. Choosing a distribution (~6-8 min)

- Fixed trials, successes $\rightarrow$ **binomial**
- Counts, no fixed upper bound $\rightarrow$ **Poisson** (or neg. binomial if overdispersed)
- Bounded proportion $\rightarrow$ **beta**
- Symmetric continuous $\rightarrow$ **normal**
- Skewed positive continuous $\rightarrow$ gamma/lognormal (named only)
- **Match the mechanism, not the histogram shape**

Discussion (if time)

- *Which distribution fits last week's detection counts (Act 2)? Body mass (Act 3)? How might reality violate that choice?*